



THREAT ADVISORY

**ATTACK
REPORT**

Enemybot malware expands its arsenal by exploiting well-known vulnerabilities

Date of Publication

2 June 2022

Last Updated

A2

TA Number

TA2022111

Summary

EnemyBot, a Mirai-based botnet, is expanding its arsenal by exploiting well-known vulnerabilities in log4j, VMware workspace, Spring Framework, and others. Keksec, also known as Nero and Freakout, is the threat actor behind EnemyBot. Since 2016, this group has been known for crypto-mining and distributed denial-of-service (DDoS) attacks.

CVEs

CVE	NAME	PATCH
CVE-2021-44228	Apache Log4j RCE vulnerability	
CVE-2021-45046	Apache Log4j denial of service vulnerability	
CVE-2022-1388	F5 BIG-IP iControl REST Authentication missing authentication vulnerability	
CVE-2020-7961	Unauthenticated RCE via JSONWS vulnerability in Liferay	
CVE-2021-4039	Zyxel OS command injection vulnerability	

CVE	NAME	PATCH
CVE-2022-22947	Spring Cloud Gateway Code Injection Vulnerability	✓
CVE-2022-22954	VMware Workspace ONE Access and Identity Manager Server-side Template Injection Remote Code Execution Vulnerability	✓
CVE-2021-36356	Unauthenticated Remote Code Execution in VIAware	✗
CVE-2021-35064	Privilege escalation in VIAware	✗
CVE-2020-5902	BIG-IP TMUI RCE vulnerability	✓



Potential MITRE ATT&CK TTPs

TA0001 Initial Access	TA0008 Lateral Movement	TA0011 Command and Control	T1190 Exploit Public-Facing Application
T1210 Exploitation of Remote Services	T1021 Remote Services	T1132 Data Encoding	T1001 Data Obfuscation
T1030 Proxy	T1030.003 Proxy: Multi-hop Proxy		

Technical Details

#1

The source code for **EnemyBot** is already available on **GitHub**. The first file in the repository is **cc7.py**, a Python script that **downloads** all dependencies and fits the malware into the appropriate **OS architectures** (x86, ARM, macOS, OpenBSD, PowerPC, MIPS). Following that, a file called **update.sh** is automatically generated to spread the malware to vulnerable targets.

#2

The second section of the code is the **main** botnet source code, which **contains** all the **other malware** functions **without** the **vulnerability exploits** and integrates code from other botnets such as **Mirai**, **Qbot**, and **Zbot**. The malware then **compiles** the third component, **hide.c**, to avoid detection. The final package enables the malware to receive **command-and-control (C&C)** capabilities and **execute** commands from the **attacker**.

#3

EnemyBot is well-known for **exploiting** one-day vulnerabilities in products such as **Apache Log4j** (CVE-2021-44228 and CVE-2021-45046), **VMware Workspace ONE** (CVE-2022-22954), and others showing that the threat group behind this malware, **Keksec**, is well-resourced and exploits **vulnerabilities** before they are **patched**.

✂ Indicator of Compromise (IOC)

TYPE	VALUE
SHA256	7c0fe3841af72d55b55bc248167665da5a9036c972acb9a9ac0a7a21db016cc6, 2abf6060c8a61d7379adfb8218b56003765c1a1e701b346556ca5d53068892a5, 7785efeeb495ab10414e1f7e4850d248eddce6be91738d515e8b90d344ed820d, 8e711f38a80a396bd4dacef1dc9ff6c8e32b9b6d37075cea2bbef6973deb9e68, 31a9c513a5292912720a4bcc6bd4918fc7afcd4a0b60ef9822f5c7bd861c19b8, 139e1b14d3062881849eb2dcfe10b96ee3acdbd1387de82e73da7d3d921ed806

TYPE	VALUE
SHA256	4bd6e530db1c7ed7610398efa249f9c236d7863b40606d779519ac4ccb89767f, 7a2a5da50e87bb413375ecf12b0be71aea4e21120c0c2447d678ef73c88b3ba0, ab203b50226f252c6b3ce2dd57b16c3a22033cd62a42076d09c9b104f67a3bc9, 70674c30ed3cf8fc1f8a2b9ecc2e15022f55ab9634d70ea3ba5e2e96cc1e00a0, f4f9252eac23bbadcbd3cf1d1cada375cb839020ccb0a4e1c49c86a07ce40e1e, 6a7242683122a3d4507bb0f0b6e7abf8acef4b5ab8ecf11c4b0ebdbded83e7aa, b63e841ded736bca23097e91f1f04d44a3f3fdd98878e9ef2a015a09950775c8, 4869c3d443bae76b20758f297eb3110e316396e17d95511483b99df5e7689fa0, cdf2c0c68b5f8f20af448142fd89f5980c9570033fe2e9793a15fdfdadac1281
IPv4	80.94.92[.]38

Patch Links

<https://logging.apache.org/log4j/2.x/security.html>

<https://support.f5.com/csp/article/K23605346>

https://portal.liferay.dev/learn/security/known-vulnerabilities/-/asset_publisher/HbL5mxmVrnXW/content/id/117954271

<https://tanzu.vmware.com/security/cve-2022-22947>

<https://www.vmware.com/security/advisories/VMSA-2022-0011.html>

<https://www.zyxel.com/support/OS-command-injection-vulnerability-of-NWA1100-NH-access-point.shtml>

<https://support.f5.com/csp/article/K52145254>

References

<https://cybersecurity.att.com/blogs/labs-research/rapidly-evolving-iot-malware-enemybot-now-targeting-content-management-system-servers>

What Next?

Book a free demo with **HivePro Uni5** to check your exposure to this advisory. HivePro Uni5 is a Threat Exposure Management Solution that proactively reduces an organization's attack surface before it gets exploited.



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2 June 2022 • 4:45 PM

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